

Traffickcam: An AI tool for global scale hotel recognition to support sex-trafficking investigations

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George Washington University





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This is a hotel room. Do U know which one?
This info can help police solve a child sexual
abuse case. Tell us on
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Hotels, Trafficking and Images

Human traffickers operate out of hotels

Traffickers advertise online with images taken inside hotel rooms

Identifying the specific hotels used can help:

- Find victims
- Document crime locations
- Understand criminal networks



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A story in two parts

- Serendipity in how we ended up working on this project.
- Illustration of the challenges of biased datasets.



New hope in cold case of decapitated girl found in St. Louis, but now where's the body?

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March 17, 2013 12:45 am • By Christine Byers cbyers@post-dispatch.com 314-340-8087 30

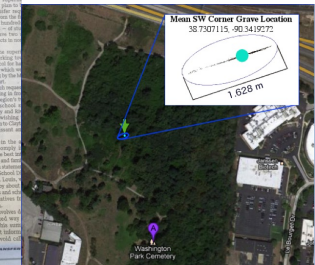


The Rev. John Heyward Jr. officiates during a funeral for an unknown girl on Dec. 2, 1983, as funeral director Jesse...

ST. LOUIS • Police could never figure out who she was. Where she came from. Or who killed her. Now, they can't even find the grave holding the decapitated remains of a child slain 30 years ago — bones that offer the remote chance of a lead in a stone-cold case that otherwise has none.

Scavengers discovered the little girl, probably between the ages of 8 and 11, on Feb. 28, 1983, decomposing in an abandoned home on the north side. Her head was never found. Despite heavy publicity, nobody identified her. Police

ei EXCHANGE INITIATIVE
Real Resources to End Sex Trafficking

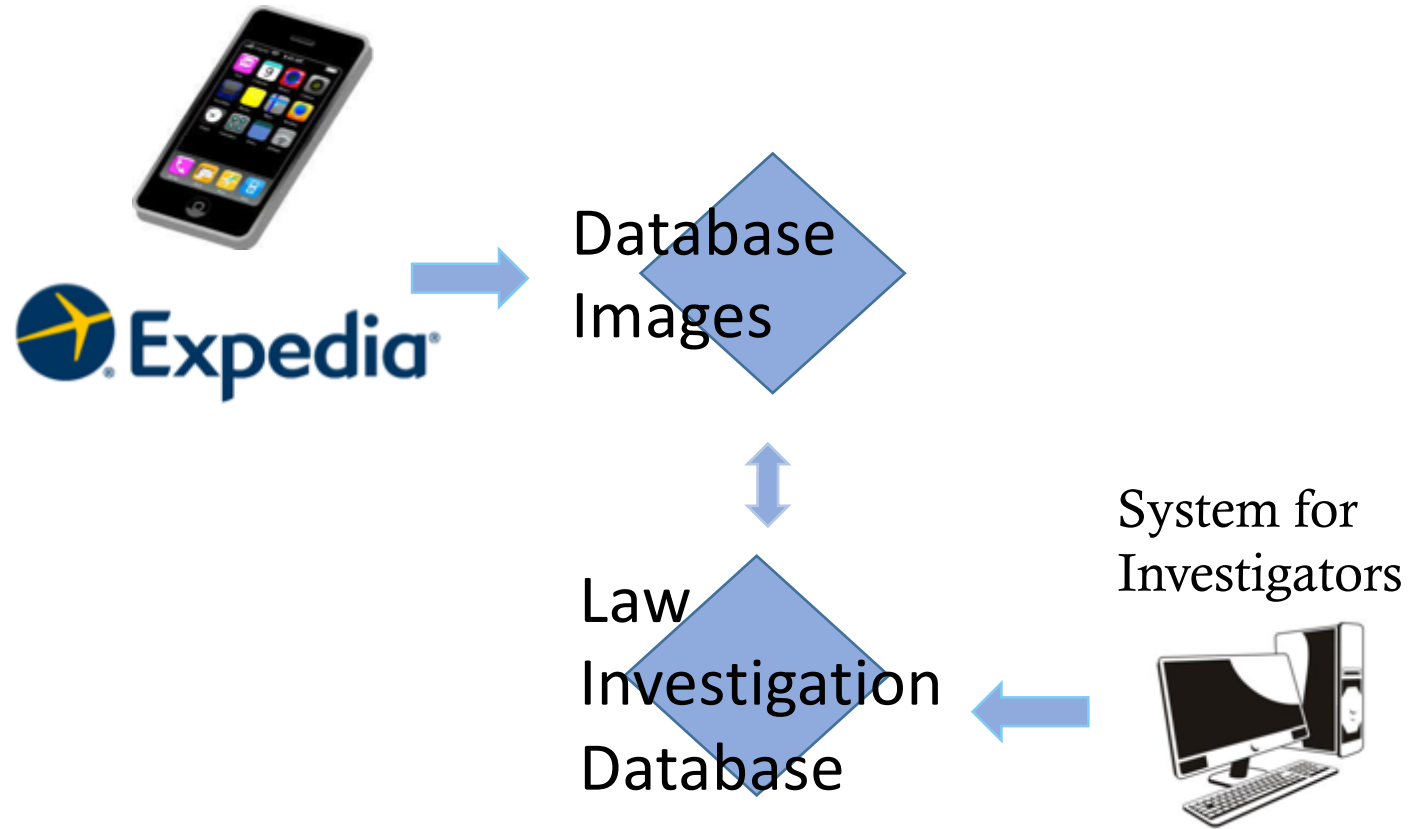


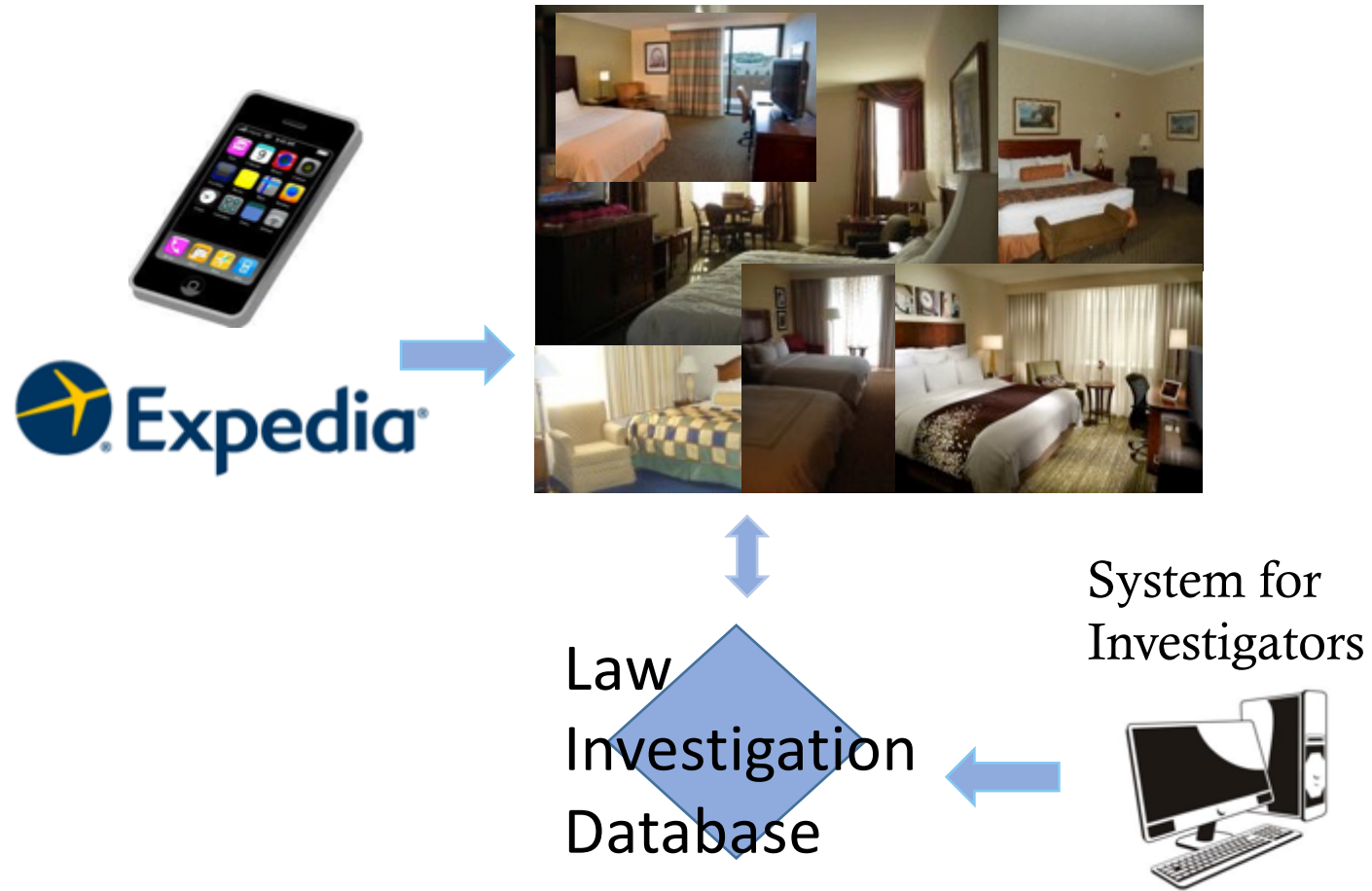


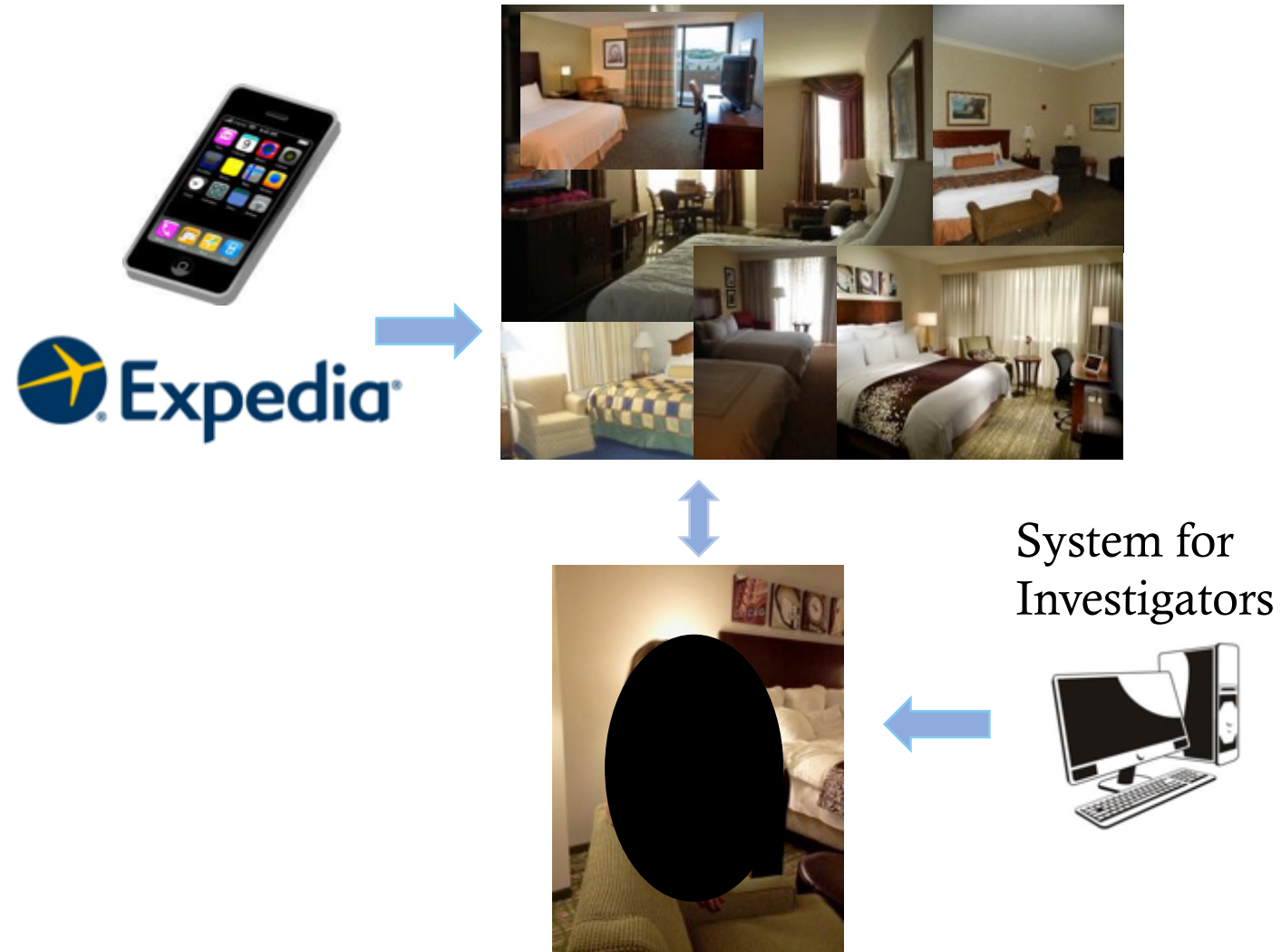
FBI
SAINT LOUIS

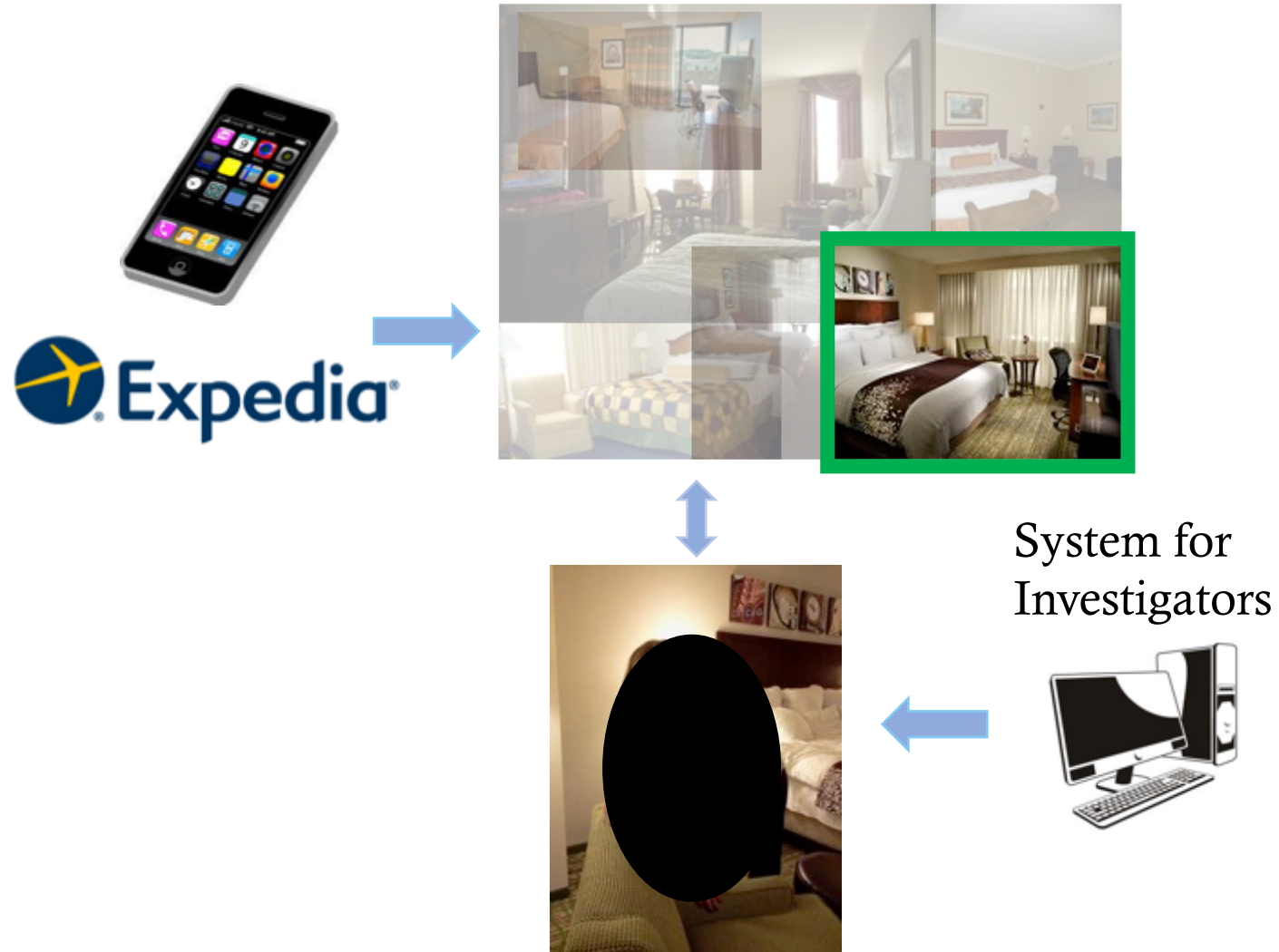
CITIZENS ACADEMY
ALUMNI ASSOCIATION
INC.

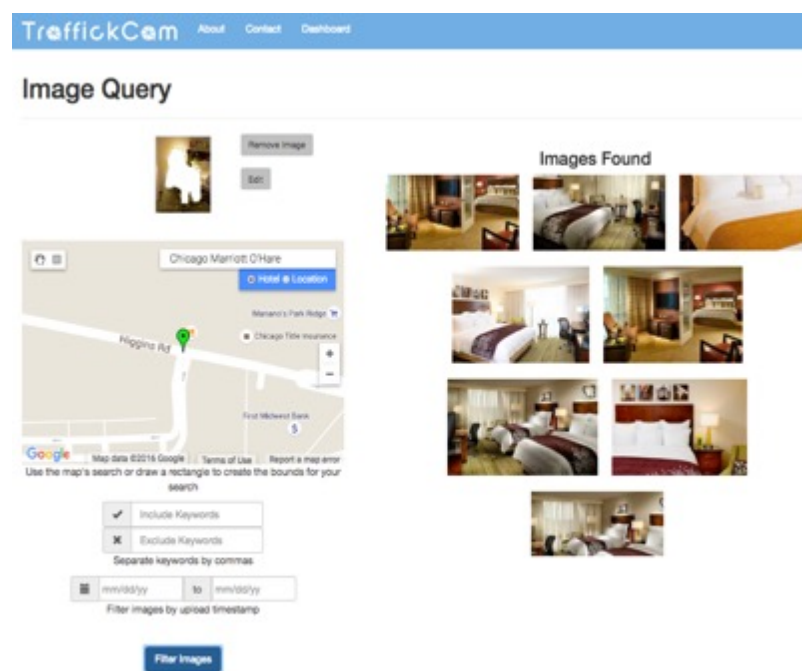
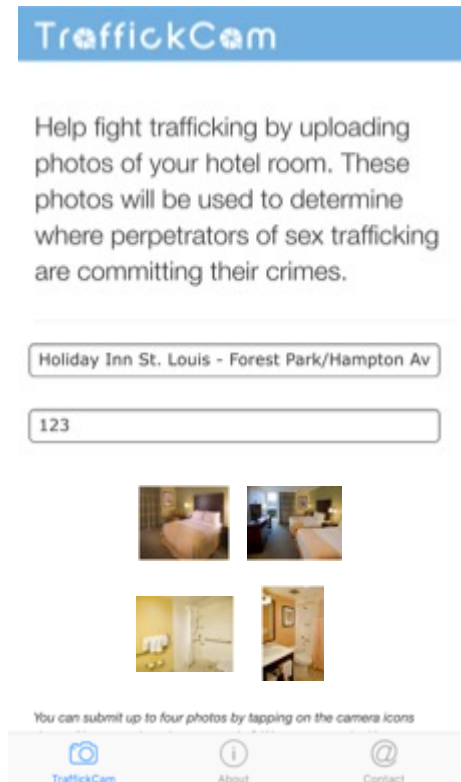
The FBISTLCAAA is affiliated with, but separate from, the FBI.









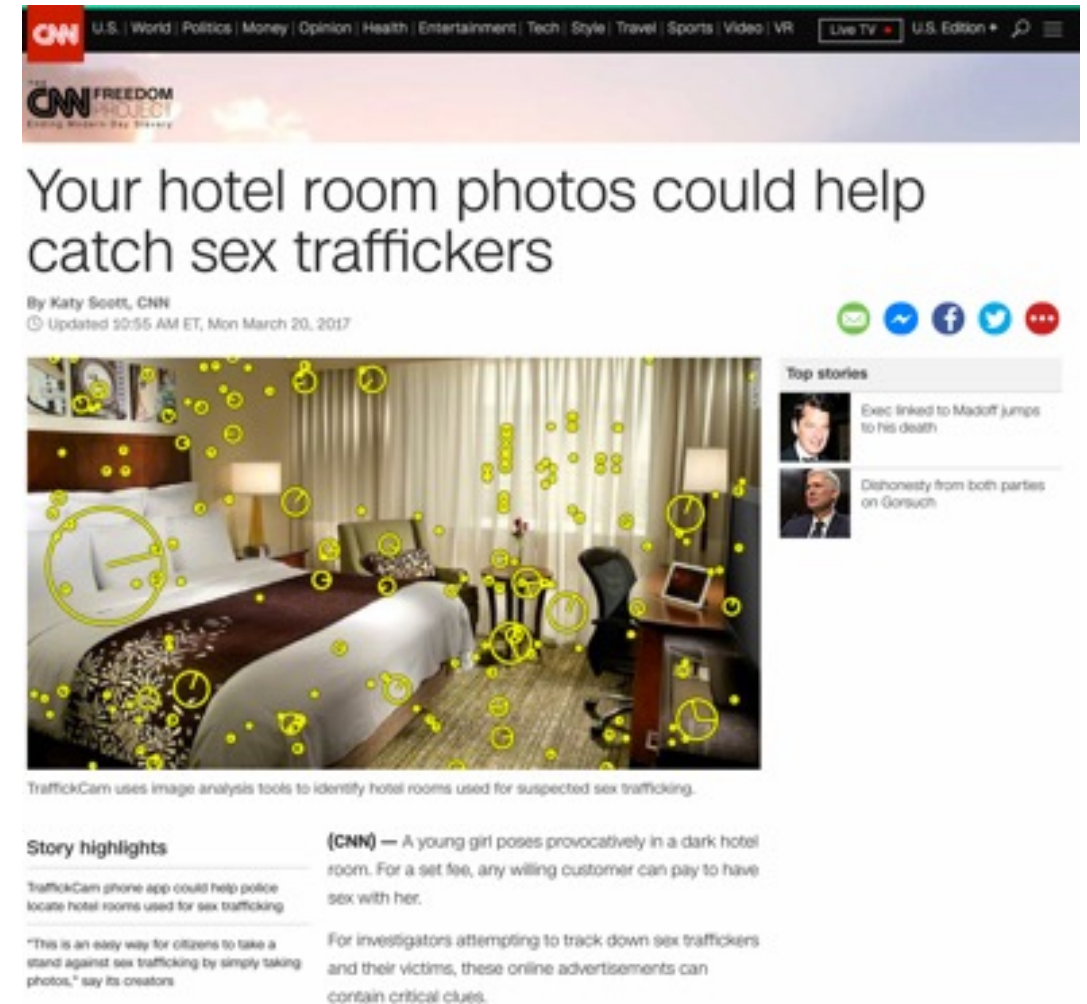


Free for iOS and Android

TraffickCam

Application Usage & Dataset Statistics

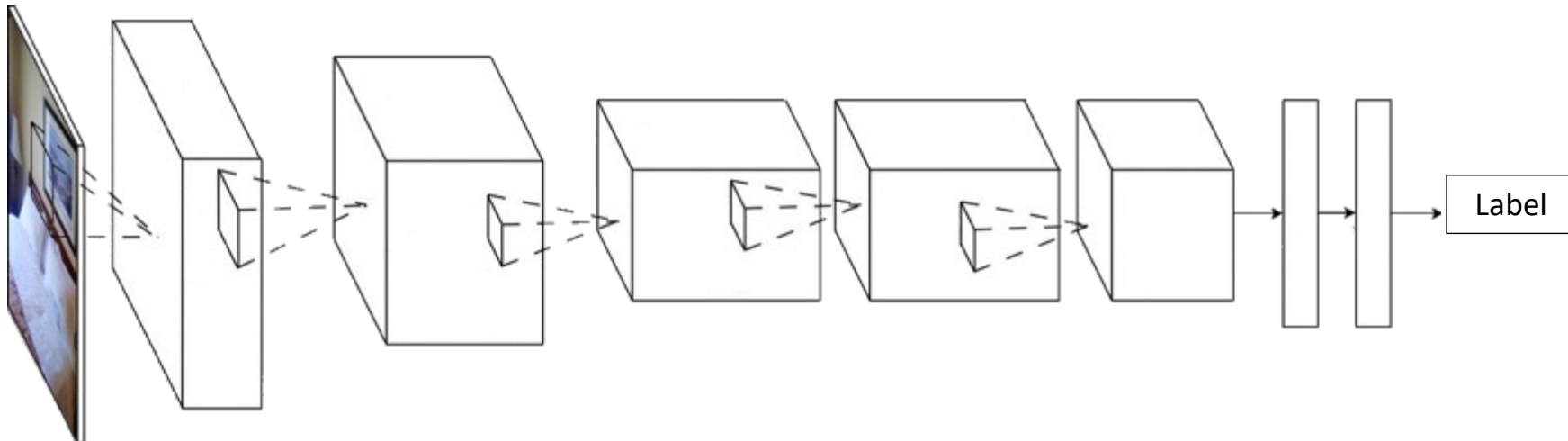
- iOS downloads: 268,700
- Android downloads: 128,500
- Total number of images: 12,792,958
- Images from TraffickCam: 1,022,953
- Unique hotels: 510,311
- Average submissions per day: 130



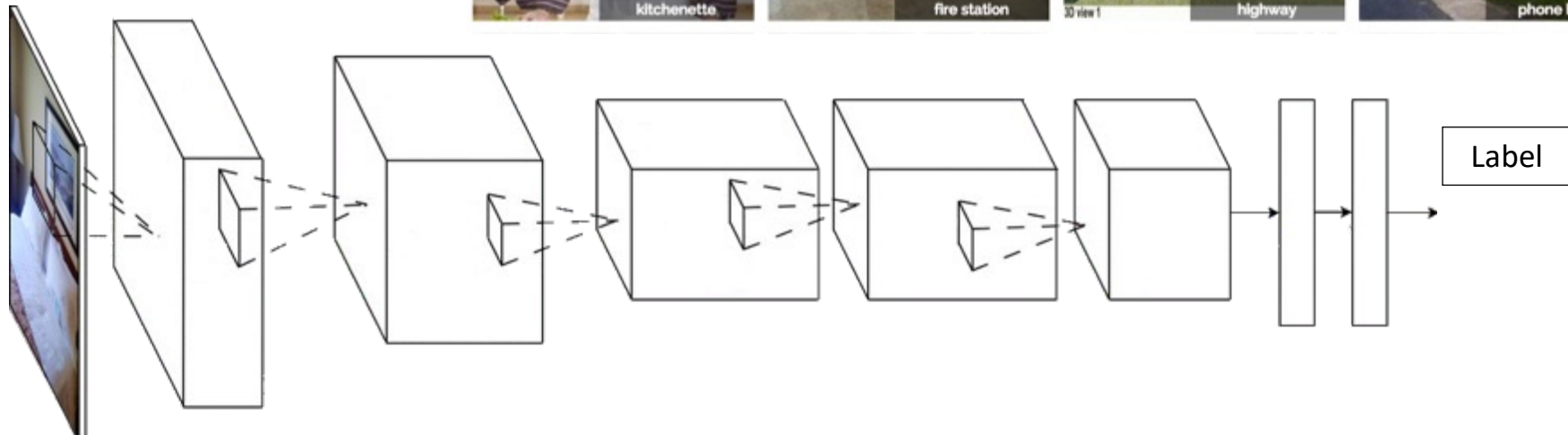
Story 2: Deep Learning, Images and Bias.

Deep Learning Features:

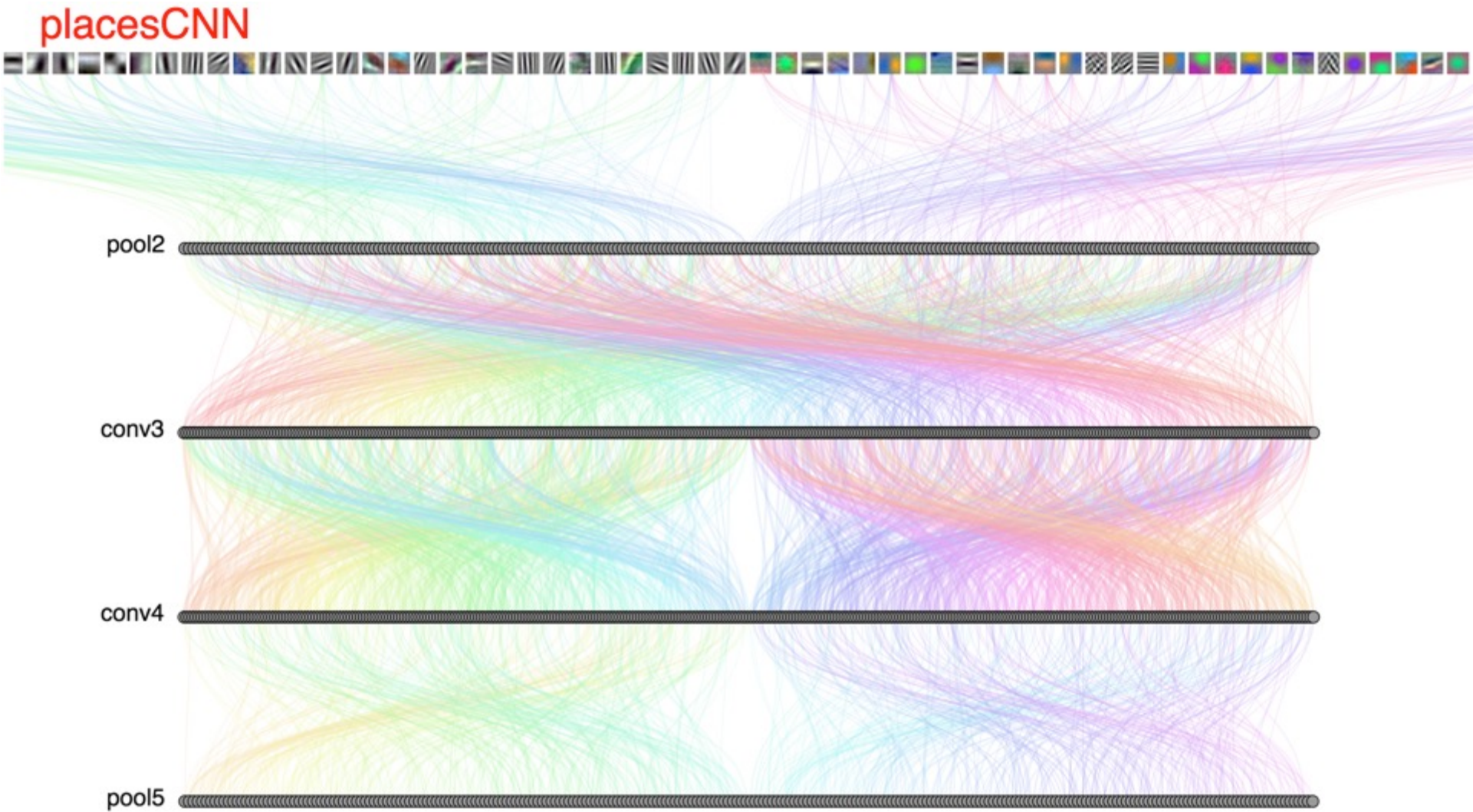
- Convolutional Neural Network architecture (Deep Dreaming, Inception, Alpha-Go)



Places dataset:
Hundreds of
categories of
different types
of places



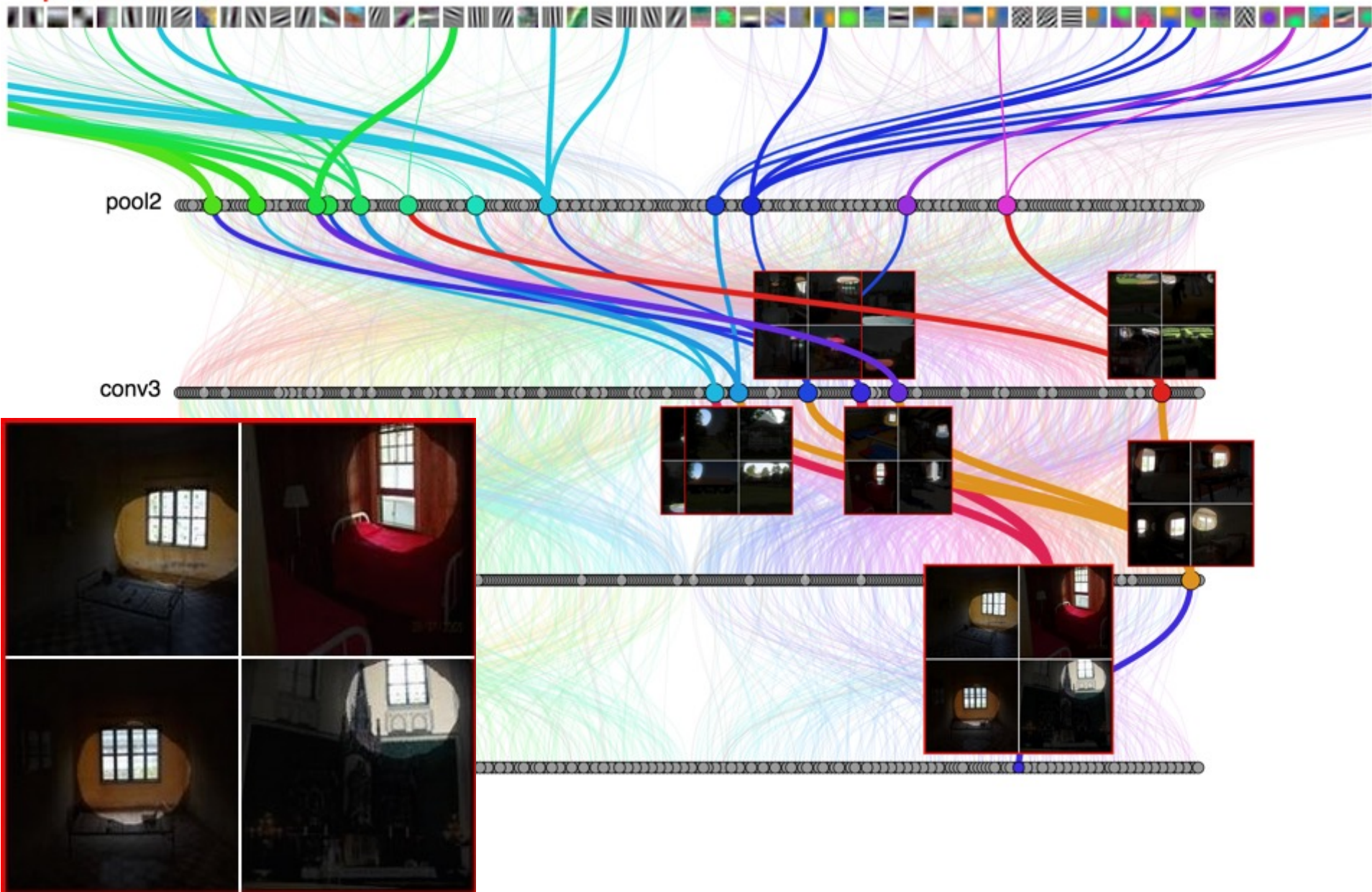
DrawNet: <http://people.csail.mit.edu/torralba/research/drawCNN/drawNet.html>



placesCNN



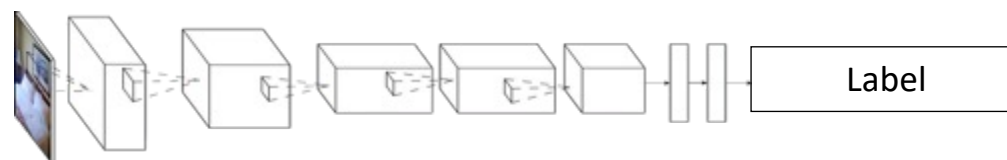
placesCNN



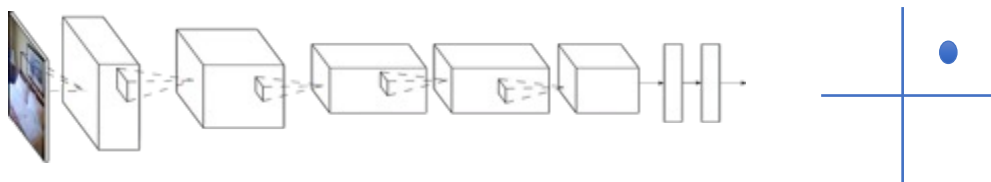
placesCNN

Amazing, but we don't need to figure out that our images are hotel rooms (we already know that),

Instead we need to know which of the 200,000 hotels the room was in



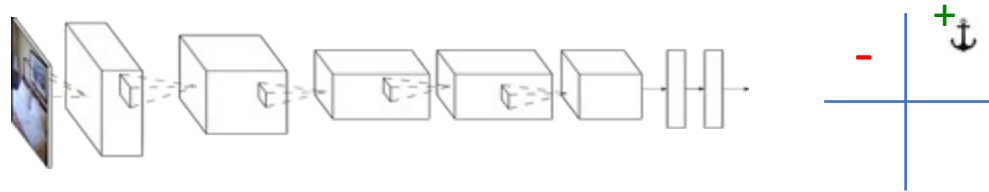
Problem: 200,000 categories... Not enough data per category



+



-



Optimize so the positive example is placed closer to the anchor point than the negative example.

This is called “contrastive (or triplet) learning”

+



...

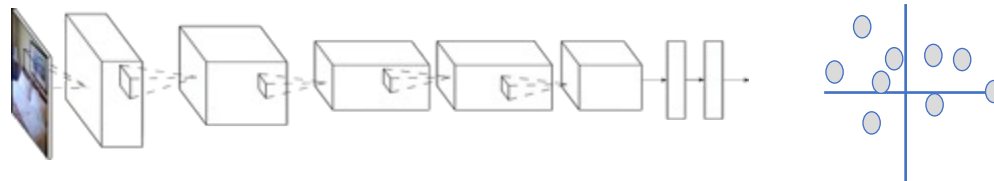
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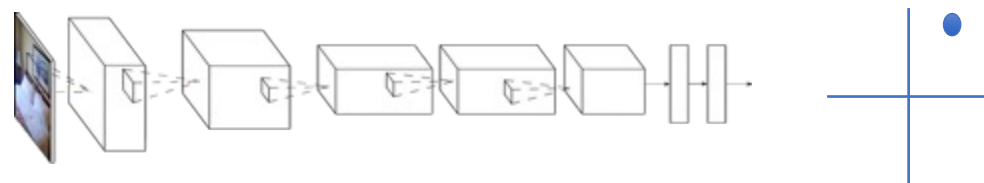
Image database

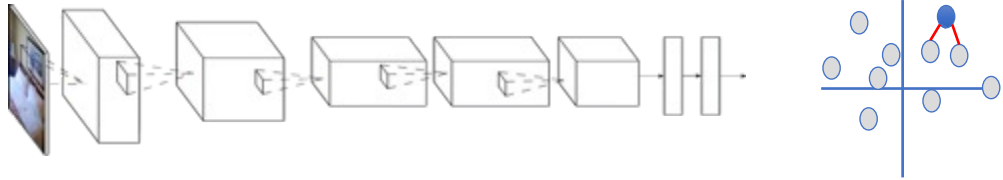


Database (in
feature space)



Query Image







places

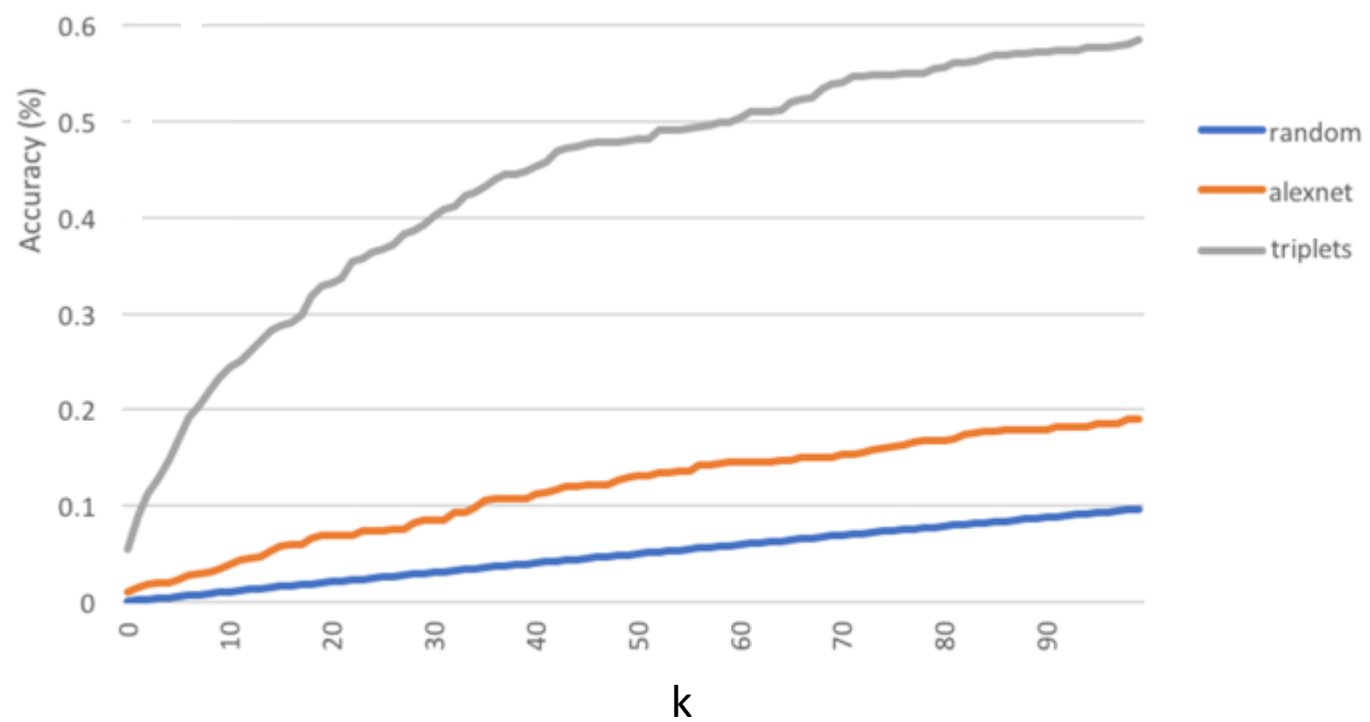
triplets



places

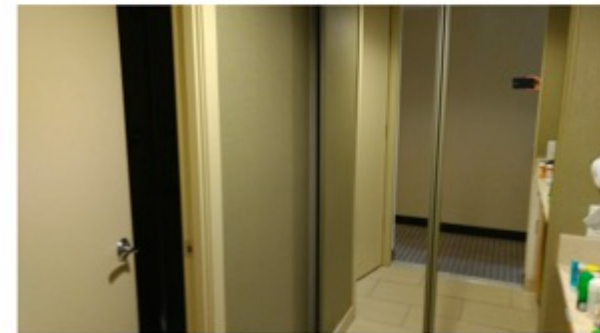
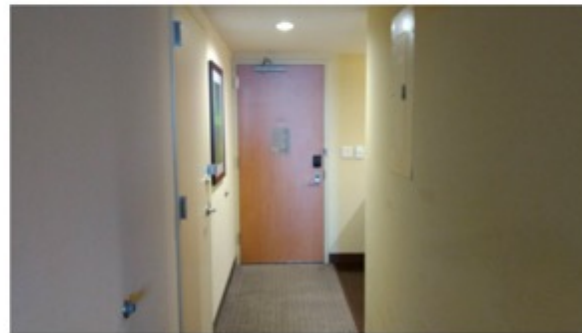
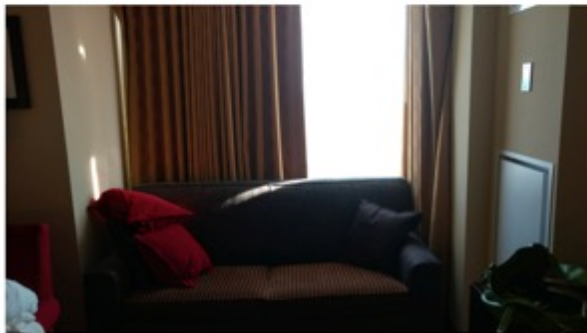
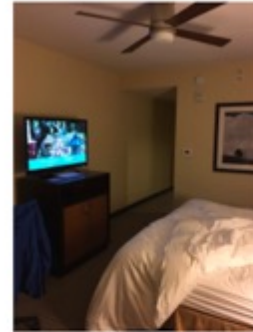
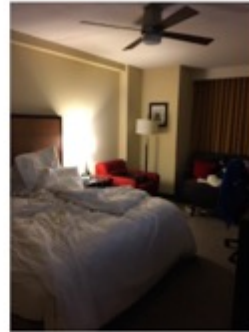
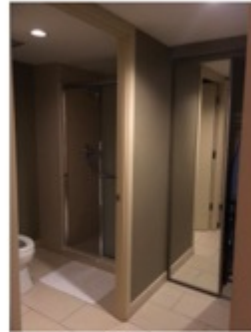
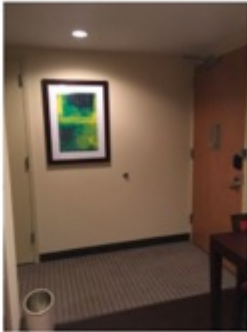
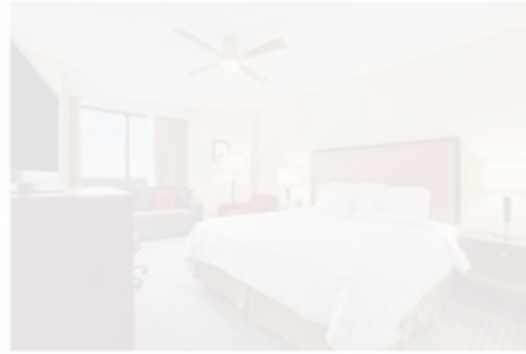
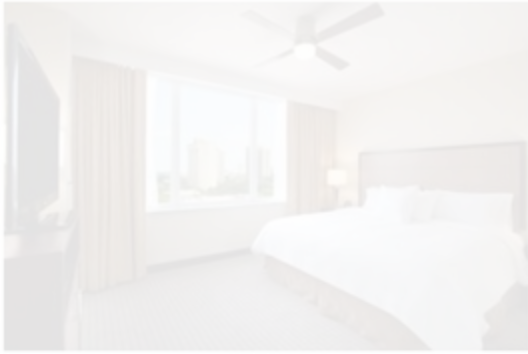
triplets

Top-k accuracy



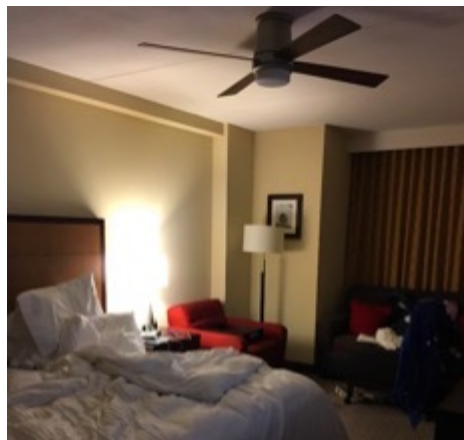
Expedia images from one hotel



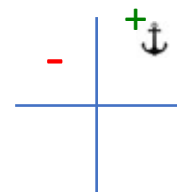
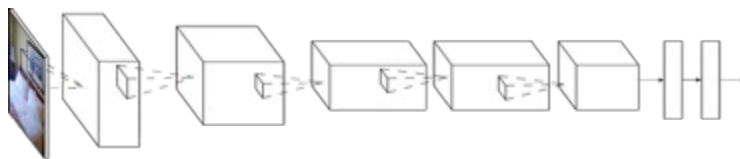
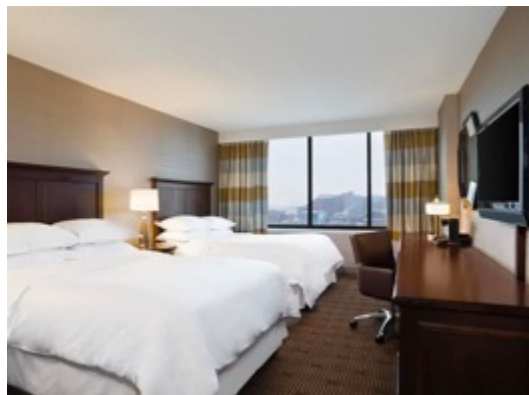


Taffickcam images from same hotel

+



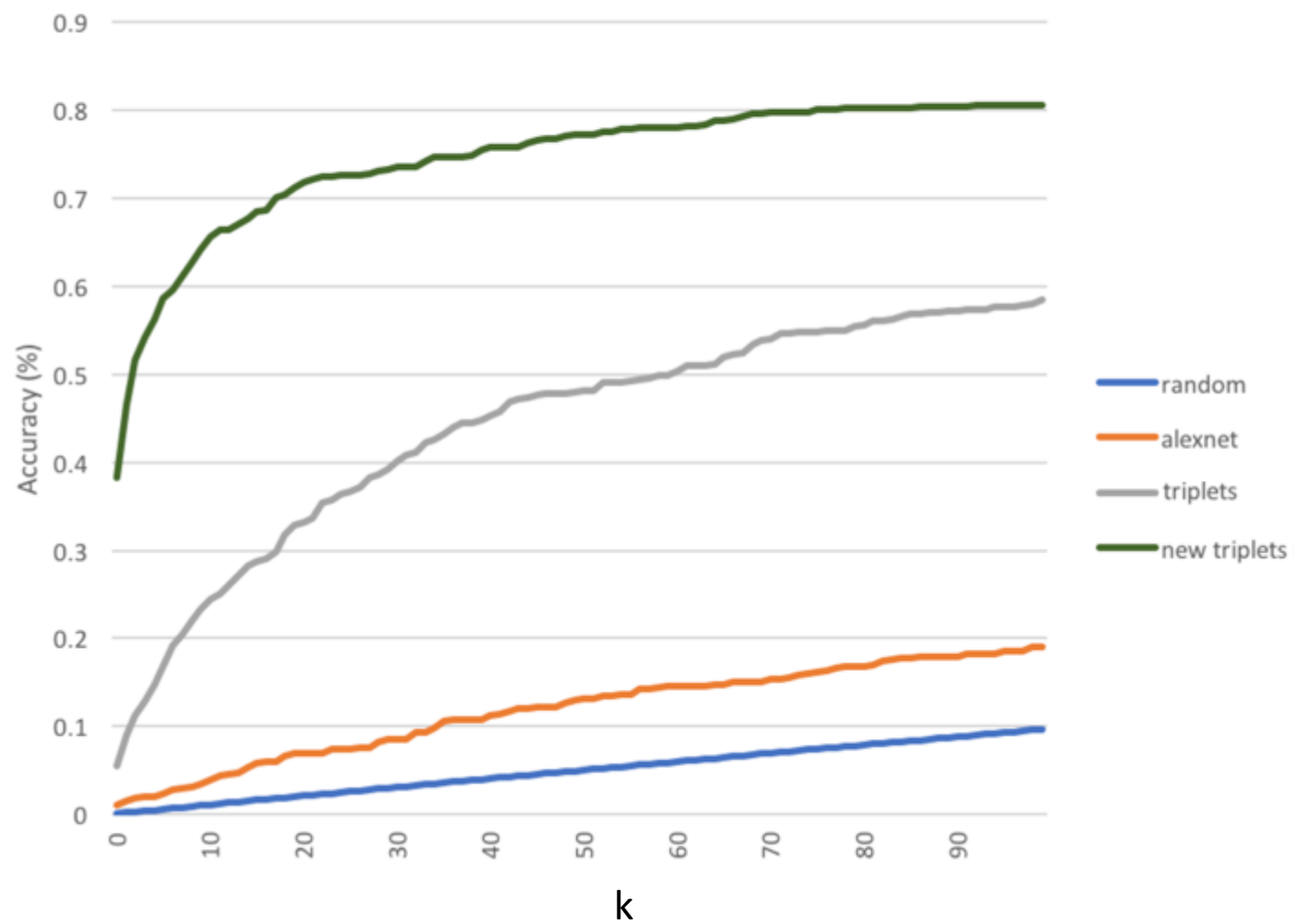
-





TraffickCam + instagram filters + adding words on images.

Top-k accuracy



Did it work?



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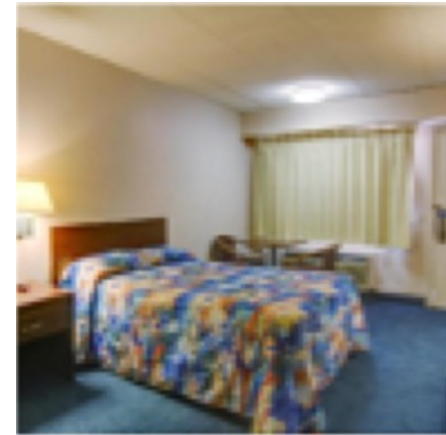


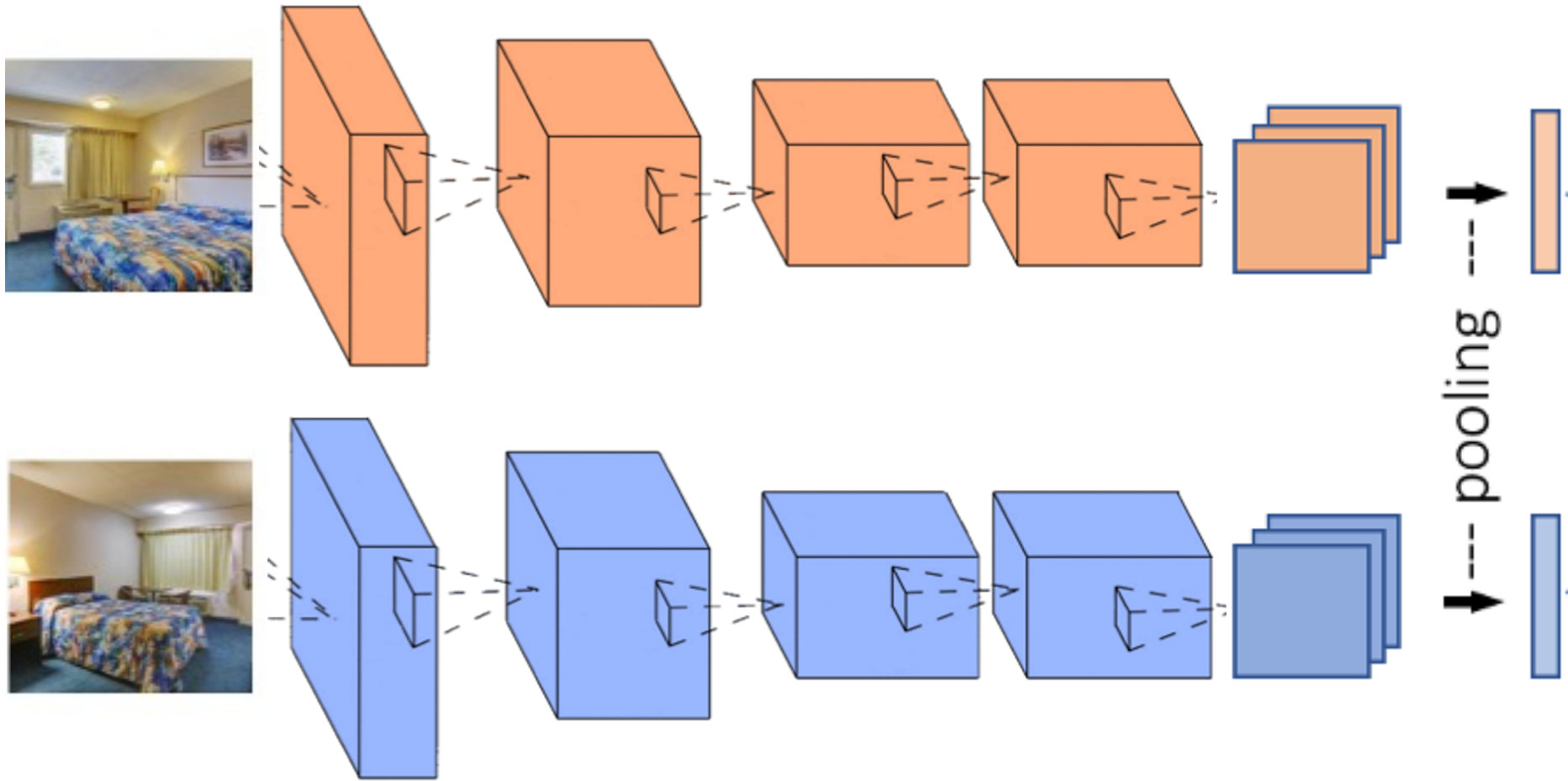
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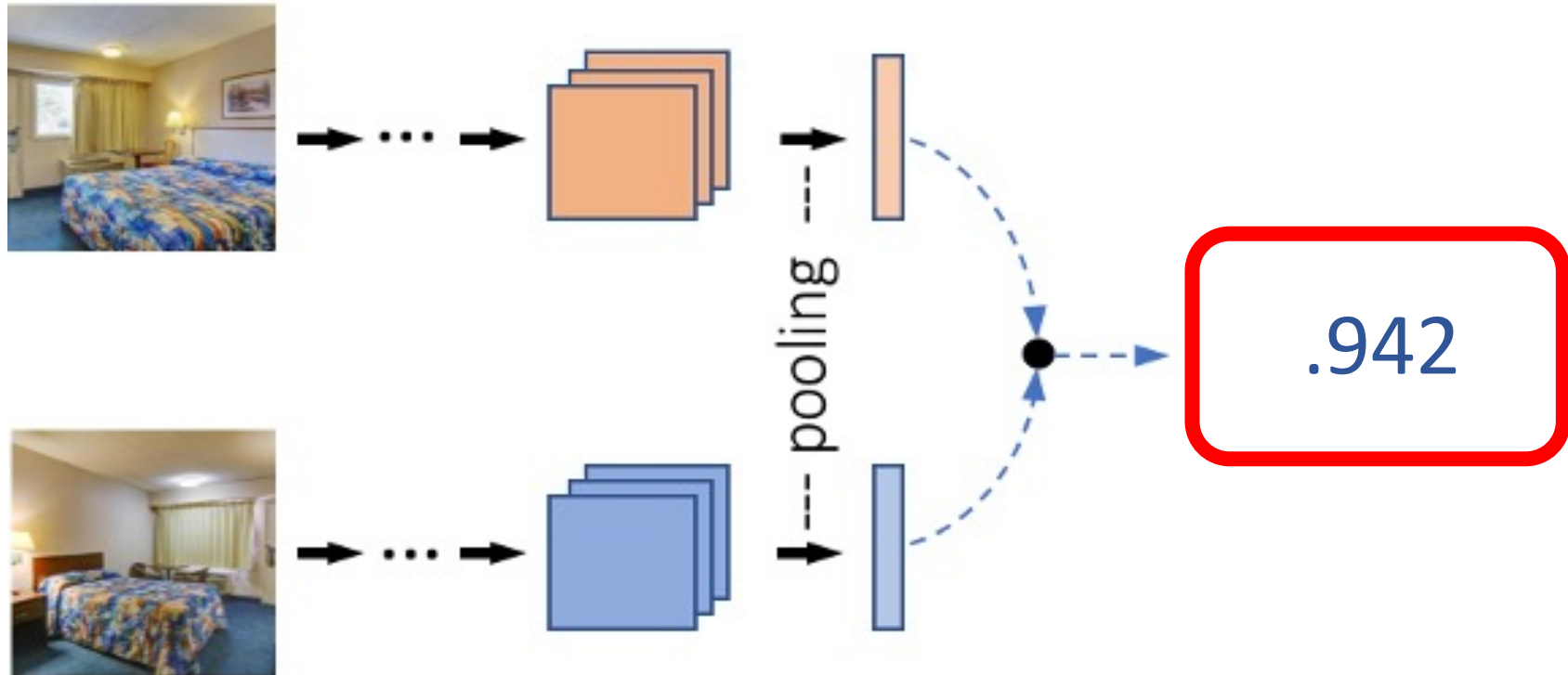
Similarity Networks

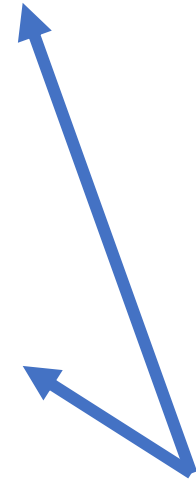


Why did the network
“think” two images
were similar?

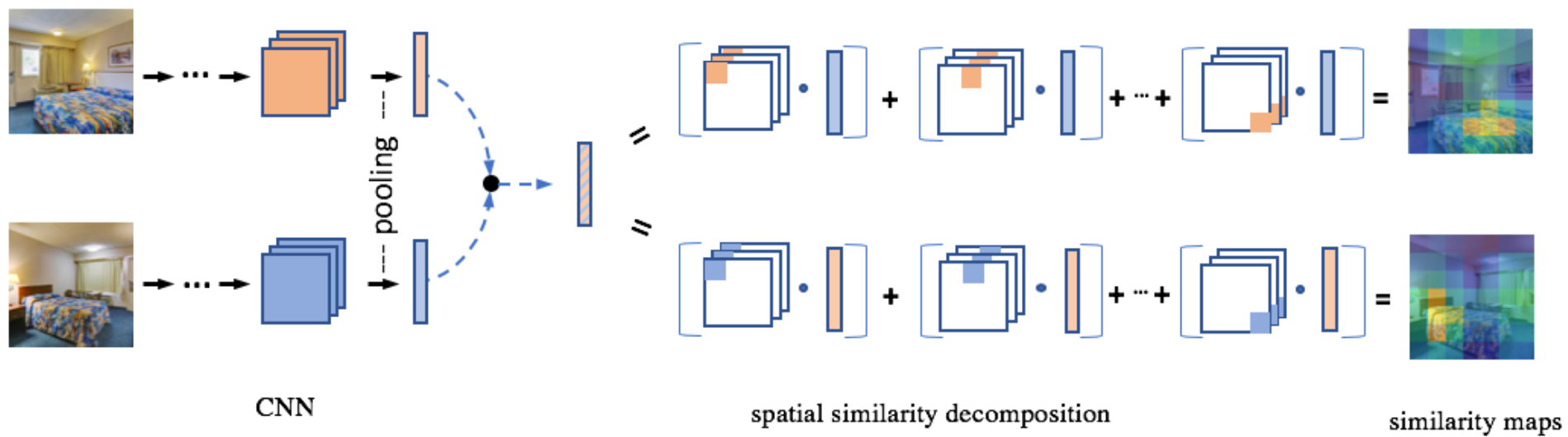






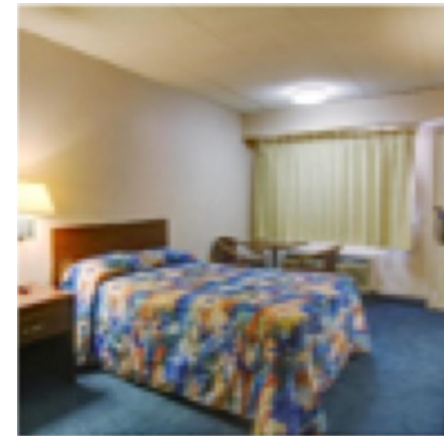


2048 7 x 7 filters





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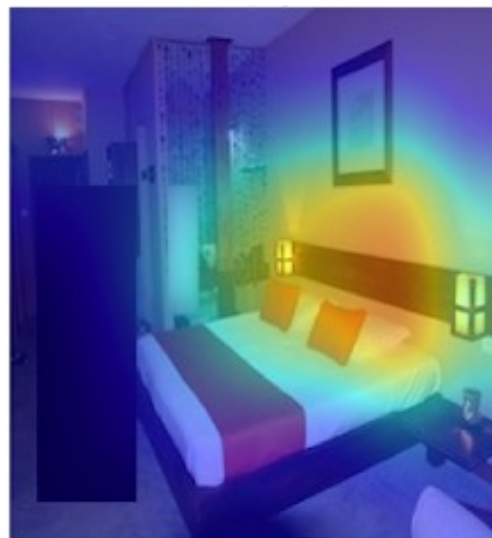


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Thank you!!

- Reach out: pless@gwu.edu